Equal Reach, Unequal Effects: Agent-Based Counterfactuals on Brazil's Protective-Order Pathway

Supplementary Information

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SI A. The SobreVIDA model description, following the ODD protocol

SI A describes SobreVIDA, an agent-based model of intimate partner violence and of Brazil's emergency protective-order pathway, following the ODD protocol in its second update (Grimm et al., 2020). It is self-contained: a reader who has not seen the paper can reimplement the model from SI A alone. Section A.1 gives the overview, including the full state-variable inventory and the within-step schedule. Section A.2 gives the design concepts. Section A.3 gives initialisation, input data and the submodels, one equation at a time. Section A.4 gives the implementation, the calibration protocol and the experimental design. Section A.5 gives a crosswalk to the ODD+D extension for human decision making (Müller et al., 2013). Throughout, the description is of the model as implemented, and every departure from a stylised reading of the mechanism is recorded as an implementation note.

A.1 Overview

A.1.1 Purpose and patterns

Purpose. SobreVIDA exists to run counterfactuals on the pathway that Brazil's Maria da Penha Law (Lei 11.340/2006) defines between harm and protection. A woman is assaulted, she reports, a *medida protetiva de urgência* (MPU, an emergency protective order) is requested and granted, and that order is enforced. Each link is administered by a different institution and recorded in a different register, so no Brazilian source observes the same woman across all four. The model's purpose is to compose the four stages over a single simulated population and then to ask what happens to the violence women live with when the chain is removed, left unenforced, extended, or entered more often.

The model is built to produce a *difference*, the same population, instantiated from the same census substrate and run under two parameter vectors, compared replicate by replicate under common random numbers. Levels are calibration moments.

The model is not intended to forecast the incidence of intimate partner violence in any Brazilian city, to rank neighbourhoods by risk, or to attribute risk to any identified individual. It is intended to compare policy configurations of the protective-order chain against one another, conditional on a specification whose assumptions are enumerated in Section A.3.

Patterns used to evaluate the model. The model is calibrated against six empirical moments on the Porto Alegre metropolitan region (Table A.1). Two patterns are held out from calibration and used as tests: repeat reporting by the same woman within a year (measurable only in the Minas Gerais police microdata, the sole source carrying a victim identifier) and the lifetime share of assaulted women who have separated. The share of femicide victims who held a prior order against the same defendant is reported with them. The shape of the within-year assault-count distribution is compared but not matched, and the failure is reported.

Table A.1: Patterns the model is evaluated against. The first block is used for calibration and the second is held out. Values obtained are the 150-replicate Porto Alegre means, except repeat reporting, which is Belo Horizonte at 40 replicates. Sources are in the paper and derivations in SI B, *Calibration targets and procedure*.

Moment	Target	Obtained
Calibration moments		
Lifetime prevalence of physical IPV	0.267 ± 0.02	0.271
Assaults per victim-year	[2.58, 3.55]	2.65
Reporting per assaulted woman-year	[0.142, 0.257]	0.185
Separation per assaulted woman-year	0.15 ± 0.03	0.161
Non-compliance per granted order	0.21 ± 0.04	0.244
Femicide per 100k (model denominator)	[2.19, 2.51]	2.33
Held out from calibration		
Repeat reporting within the year (Belo Horizonte)	1.163 per reporting woman	1.376, too many
Share no longer with the aggressor	0.79 (DataSenado)	0.390, beyond a 12-year run
Femicides with a prior order (CNJ)	0.324	not compared
Within-year assault-count shape	FBSP four bands	not matched

A.1.2 Entities, state variables, and scales

The model contains three kinds of entity: *persons*, of two subtypes, the *relationship* that joins two persons, and the *model* itself, which carries the calendar, the institutional registers and the spatial environment. There is no separate household, firm, court, or police entity: the institutions of the protective-order chain enter as rates and as state carried by the aggressor, not as agents.

Person. A person is an individual man or woman drawn from census microdata. The base class carries the socio-demographic attributes, the stress stock, the disposition, and the inhibition threshold. Two subclasses specialise it. *Men* carry the aggression side of the model: the violence trigger, the escalation offset, and every piece of protective-order state. *Women* carry the victim side: a second disposition governing the response to an assault, and its own threshold. Everything upstream of action (stress, the societal term, peer influence, disposition) is computed identically for both, so the gender asymmetry in the model is confined to the single step at which disposition is turned into behaviour.

Relationship. A relationship is a first-class entity rather than a link attribute. It holds the two partners, the date it began, and a flag recording that it has ended. Escalation and lethality accumulate in it and are cleared when it is released. A relationship that has held violence outlives the end of cohabitation by a one-year post-separation window (Section A.3.3.8). One that has held none ends at the breakup.

Environment. The environment is threefold. Spatially, each agent carries the code of its *área de ponderação* (AREAP), the finest unit at which Brazilian census microdata are disclosed, and through it the code of its municipality. Every model output can be disaggregated to that geography. Temporally, the model carries a calendar date, which indexes the municipal homicide series. Institutionally, the model carries two registers of removal from the population, custody

after a femicide, counted in years, and custody after an arrest in flagrante for breaching an order, counted in months. It also carries order-level counters used for measurement.

State variables. Table A.2 is the full inventory. Variables marked *instrumentation* are read out of the model and never read back into it: removing them changes no trajectory.

Table A.2: State variables by entity. σ, δ, τ, δ^A and τ^A are the five quantities that carry the model. Everything else is either a fixed attribute, a counter or institutional state.

Variable	Type	Meaning
Person: attributes fixed at creation		
`gender`	string	male or female, decides the subclass
`age`	integer	years, incremented on the agent's own birth month
`color`	category	census race category (*branca, preta, parda, amarela, indígena*)
`years_study`	integer	years of schooling, constrained to be attainable at the agent's age
`address`	integer	AREAP code, whose first seven digits are the municipality
`house_location`	category	urban or rural
`birth_month`	integer	month in which the agent ages
τ_i `threshold`	real	inhibition threshold, $\sim \mathcal{U}[\tau_{\min}, \tau_{\max}]$, drawn once, mutable thereafter only through escalation and protective orders
Person: attributes that change		
`is_working`	boolean	employment status, switches with probability `CHANCE_CHANGE_WORKING_STATUS` per step
`work_modality`	category	formal or informal, drawn at creation for those employed, `None` otherwise
`wage`, `reserve_wage`	real	individual income, normalised to $[0, 1]$ across the population
`num_members_family`	integer	size of the relationship the agent belongs to
σ_i `accumulated_stress`	real	stress stock, non-negative
c_i `societal`	real	societal channel in $[0, 1]$
`community_violence`	real	homicides per 100k in the agent's municipality, current year
δ_i `disposition`	real	disposition to act
`weight_vector`	dict	peer network: $\lvert N \rvert$ same-gender agents with weights w_{ij}
`spouse`	ref	current partner, or `None`
`relationship`	ref	current relationship object, or `None`
`ex_partner`	ref	post-separation tie, set only when the ended relationship held violence
`months_since_separation`	integer	position within the post-separation window
`n_attacks`, `assaulted`	integer	assaults committed, lifetime, never reset. The stress inflow reads `assaulted`
`n_received_attacks`	integer	assaults received *in the current relationship*, cleared with the relationship, not at the breakup
`got_attacked`	integer	assaults received this year, reset at the year boundary
`denounce`	boolean	reported at least once this year, read by peers
`category`	string	person, aggressor, or victim
`is_dead`, `femicide_victim`	boolean	removal from the population, and its cause
`number_of_past_-relationships`	integer	instrumentation
Men only		
`threshold_escalation`	real	cumulative erosion of τ within the current relationship, capped at $\varepsilon_{\max}$
`has_mpu`	boolean	an order is in force

continued on the next page

Table A.2: *(continued)*

Variable	Type	Meaning
`mpu_time`	integer	months left on the order's own clock
`threshold_mpu_boost`	real	how much τ was raised, kept separate from the escalation offset so the two unwind independently
`mpu_patrolled`	boolean	the granted order was forwarded to a patrol that reached him
`is_monitored`, `monitoring_time`, `monitoring_level`	mixed	the visiting spell and its intensity
`mpu_violation_count`, `times_arrested`	integer	breaches and arrests in flagrante
`ever_had_mpu`, `mpu_this_relationship`	boolean	instrumentation for femicide composition
`violence_recidivism_risk`	real	instrumentation
Women only		
δ_i^A `disposition_action`	real	propensity to act on being assaulted
τ_i^A `threshold_action`	real	her own action threshold, $\sim \mathcal{U}[\tau^A_{\min}, \tau^A_{\max}]$, drawn once and fixed
`attacked_this_step`	boolean	gate on the decision stage, set by the aggressor, cleared once she has responded
`action_times`, `denounce_times`, `break_times`, `remain_times`	integer	annual counters, the last two read by peers
`ever_attacked`, `ever_broke`, `ever_separated_from_aggressor`	boolean	lifetime flags, never reset
`assaults_lifetime`	integer	lifetime count
Relationship		
`members`	dict	the two partners
`init_date`	date	when it began
`it_ended`	boolean	set by the victim's Separate action or by the removal provision of a breached order
`address`	integer	AREAP, inherited from the first member added
Model		
`current_date`	date	advances one month per step
`incarcerated`	dict	men removed after a femicide, in years
`detained`	dict	men in custody after an arrest in flagrante, in months
`mun_population`	dict	census population by municipality, the rate denominator
`violence_indice`	dict	municipal homicide counts by year
order-level counters	integer	requested, granted, patrolled, breached, breach channel, breach delay (instrumentation)
femicide counters	integer	total, with an aggressor who ever held an order, with one held in that relationship, post-separation (instrumentation)

Scales. One time step is one calendar month. The horizon is $T = 144$ steps, twelve years, running from January of the census year (2010) to December 2021. The municipal homicide series that feeds the societal channel ends in 2021, and the model cannot be stepped past it.

Spatially, the extent is a Brazilian metropolitan region (Porto Alegre for calibration, Belo Horizonte for out-of-sample transfer), and the resolution is the AREAP, of which the Porto Alegre region has 167 and Belo Horizonte 177. The model is run at a scale of 1,000 initial relationships, which after the census sampling described in Section A.3.1 yields about 1,850 adult agents (995 men and 849 women at the start of a Porto Alegre run). The calibration holds only at this scale,

since prevalence is not invariant to it.

Implementation note (space). Space in this model is *attributive* rather than *relational*: an agent carries a location, and no process resolves anything through adjacency. The AREAP enters twice. It is the sampling unit for the population, so an agent's age, gender, race, schooling and wage are drawn from the census distributions of the area she lives in, and pairing is within-area (§A.3.1). Its first seven digits give the municipality, which carries the homicide series feeding the societal channel (§A.3.3.3). Peer networks are sampled from the whole population rather than by proximity, and no term reads the distance between two agents. The code also instantiates a spatial grid, which no process reads (§A.4.2).

A.1.3 Process overview and scheduling

Within a month the model executes the following, in this order.

1. **Release from custody.** Men whose detention term has expired are returned to the schedule, before anyone acts.

2. **Agent activation, by class.** Classes are activated one at a time (relationships, then men, then women), and within each class the order is reshuffled every month. Relationships have no step of their own. Every person, of either gender, runs the same five-part sequence:

   (a) An age-dependent draw of natural death, which if it succeeds ends the step.

   (b) A Poisson call that puts the agent on the accumulating or the relief branch of the stress stock, and updates σ accordingly (Section A.3.3.2). Employment and wage may change first.

   (c) The societal term c (Section A.3.3.3).

   (d) The disposition δ, by Rescorla–Wagner updating towards a target assembled from four channels (Section A.3.3.5).

   (e) *Act*, which is the only gender-specific step.

   A partnered adult man evaluates the violence trigger. A man with no partner but an open post-separation tie evaluates the pursuit of his ex-partner. A woman updates her propensity to act and, *if she was assaulted this month*, evaluates Report / Separate / Remain. After acting, a man runs down the order's clock and then the visiting spell.

3. **Data collection.** Model-level reporters are read.

4. **Population update.** Deaths are removed. A man whose partner was killed enters the incarceration register. Single agents are matched into new relationships with a Poisson-derived probability. Agents whose birth month has come age by one year, and each agent's community-violence value is refreshed for the current calendar year.

5. **Relationship update.** Relationships flagged as ended, or containing a dead member, are dissolved, which unassigns the spouses. If the relationship held violence, it opens the post-separation tie.

6. **Calendar.** The date advances one month.

At each twelve-month boundary the harness reads the annual victim-side moments, registers the AREAP and municipal panel, reviews the incarceration register, and resets the annual counters. The reset changes behaviour, because the peer action norm of Equation (A.14) reads those counters, so the norm always refers to the current year.

Two consequences of this ordering are needed to reimplement the model faithfully. First, the peer dispositions an agent reads are a mixture of month-t and month-$(t-1)$ values, depending on where its peers fell in the shuffle. The model updates asynchronously and does not double-buffer. Second, because men are activated before women, an assault and the response to it fall in the same month, the decision stage reading a flag set earlier in the same step. An assault landing after the woman has already run her own step sets the flag for the following month rather than being lost.

A.2 Design concepts

A.2.1 Basic principles

Four bodies of theory are built into the model, each at an identifiable place.

The *architecture* follows the ecological framework of Heise (1998), in which violence against women is the product of an interplay among nested levels rather than of any single cause. The four levels (individual, relationship, community, society) enter as the four channels that converge on the disposition in Equation (A.7).

The *decision architecture* is Epstein's Agent Zero (Epstein, 2014), which specifies action as the joint product of an affective, a social and a deliberative component compared against a threshold. Both sides of the model instantiate it: the aggressor's disposition against his inhibition threshold, and the victim's propensity to act against her own. The affective module's updating rule is the associative-learning equation of Rescorla and Wagner (1972), used here in both stocks. The antecedent that stacks the same three components (Agent Zero, Rescorla–Wagner and a threshold) is Ardiles Cruz (2019).

The *stress submodel* takes its shape from Drigo et al. (2012), whose specification makes couple-level violence depend on household income, the woman's income, relationship characteristics, race, adherence to traditional gender roles and informal support. SobreVIDA keeps that shape and replaces the fitted logistic with severities elicited from domain specialists in the VIDA model (Mori Madeira et al., 2021).

The *relational dynamics* come from the clinical and daily-diary literature on intimate partner violence: the cycle of violence and its conciliatory phase (Katerndahl et al., 2021a; Guidi et al., 2016), escalation as a consequence of having offended rather than of the victim's reaction (Katerndahl et al., 2021a; Gnisci and Pace, 2016), the woman's own affective state as a predictor of her partner's aggression (Katerndahl et al., 2022), the nonlinearity of her need for action (Katerndahl et al., 2021b), and the elevation of risk after separation (Campbell et al., 2007; Serran and Firestone, 2004; Campos and Colman, 2024).

The *institutional layer* is built on the finding that issuance alone is insufficient and that effectiveness depends on enforcement, monitoring and inter-institutional coordination (Strand,

2012; Dowling et al., 2018; Russell, 2012). Every link of the chain in Section A.3.3.11 is therefore a separate institution with its own coverage and can be switched off on its own.

A.2.2 Empirical warrant

For every term of Section A.3.3, Table A.3 names the empirical claim the term rests on and the source of that claim. The sign and the presence of each term are answerable to a source, and the magnitudes are modelling choices, tabulated in Table A.4 and probed in SI B, Section B.3.

Three rows carry a weaker warrant than the rest and are marked with a dagger: the race term, whose source documents a differential in outcomes rather than a mechanism behind it, the dyadic channel, implemented one way and unsigned where the evidence is two-way and emotion-specific, and the peer action norm, implemented in the direction that raises help-seeking where the same source also reports the opposite effect.

Table A.3: The empirical warrant for each mechanism of Section A.3.3. Magnitudes are not here but in Table A.4. A dagger marks the three mechanisms whose source establishes a weaker claim than the implementation makes.

Mechanism	What the source establishes	Source
Stress inflow		
Wage, negative	IPV prevalence higher among couples of lower socioeconomic status, surviving country by country across the EU	Reichel (2017)
Schooling deficit	Lower attainment associated with vulnerability to IPV and with greater public justification of it	Waltermaurer (2012)
Partner's race†	Race differential in Brazilian femicide mortality. The differential, not a mechanism behind it	Monteiro et al. (2021)
Bundle of covariates	Couple-level violence driven jointly by income, relationship characteristics, race and gender-role adherence	Drigo et al. (2012)
Severities, gender and age weights	Elicited from domain specialists on a three-level scale	Mori Madeira et al. (2021)
Environment		
Machismo index	Brazilian attitudes to gender roles by age and schooling band	Venturi and Godinho (2013)
Ambient violence	Municipal homicide rate to 2021	Brasil. Ministério da Saúde. DATASUS (2023)
Same-gender networks	IPV norm measurement defines reference groups as same-gender peers rather than the population	Seff (2021)
Aggressor		
Learning plus threshold	Agent Zero: action as the joint product of affect, social influence and deliberation	Epstein (2014), Rescorla and Wagner (1972)

Mechanism	What the source establishes	Source
Dyadic channel†	Daily diaries: her anger maximised his violence. Implemented here one way and unsigned	Katerndahl et al. (2022)
Escalation on his own inhibition	Escalation contingent on having begun to offend rather than on the victim's reaction	Katerndahl et al. (2021a), Gnisci and Pace (2016)
Post-separation window	Greatest risk after separation, concentrated in the first months, a year in the Brazilian data	Campbell et al. (2007), Serran and Firestone (2004), Campos and Colman (2024)
Lethality in prior assaults	Repetition among the characteristics of notified cases associated with the woman's death	Pinto et al. (2021)
Victim		
Propensity plus own threshold	Action depends nonlinearly on need-for-action and on prior experience of acting	Katerndahl et al. (2021b)
Autonomy as employment	Help-seeking higher among working women across eighteen DHS countries	Aboagye et al. (2023)
Peer action norm†	Social support raises help-seeking and also makes staying easier. Only the first is implemented	Katerndahl et al. (2013)
Decision gated on the incident	Help-seeking concentrated in a window after the assault, before the conciliatory phase	Guidi et al. (2016)
Honeymoon reduction	The "Calm, Loving Respite" pulls the victim back into the relationship	Katerndahl et al. (2021a)

A.2.3 Emergence

The model generates the level and dispersion of disposition across the population, the share of men ever violent, the schooling gradient in violence and the spatial pattern of all of these across AREAPs. It generates the concentration of non-compliance in high-risk offenders, which no parameter indexes to any risk profile, and the composition of femicide by whether the aggressor had held a prior order. The paper reports as findings only the paired differences between scenarios.

Lifetime prevalence and the annual rates are calibration moments. The attrition funnel is a synthesis: its first rate, reporting, is simulated, and the request and grant rates below it are entered as data and never adjusted to improve fit. No institution follows the same woman across the four stages, so their product over one population appears in no source.

A.2.4 Adaptation

Agents adapt in the weak sense of ODD: their behaviour changes with their state, but no agent evaluates alternatives or seeks an outcome.

An aggressor's propensity to assault tracks his accumulated circumstances through disposition, and his inhibition erodes with his own history of offending. A woman's propensity to act tracks her accumulated victimisation, what her peers have done, and her own economic autonomy. Both are *indirect* objective-seeking in ODD's sense at most: the agent responds to state,

and the response happens to have consequences the agent does not represent.

Two further adaptations are institutional. A man under a granted order faces a raised threshold, and if the order was visited, a monthly multiplicative suppression of disposition and stress. Neither is chosen by him.

A.2.5 Objectives

There is no objective function anywhere in the model, and no agent maximises, satisfices, or compares alternatives. Both decision stages are threshold crossings: the quantity that has accumulated is compared against a bar drawn once at agent creation, and the comparison decides the act. Remain is the default. A woman stays while her accumulated experience, her peers' example and her economic position have not carried her over her own threshold.

Without an objective function, action is driven by affect and social contagion with no utility to reconcile them, which makes the model Agent Zero rather than rational choice.

A.2.6 Learning

Both stocks learn, and by the same rule. Disposition and the propensity to act each move a fraction α of the way towards their current target each month (Equation (A.8)), so neither tracks contemporaneous conditions. The learning rate sets the memory of the behavioural state, and is why a man whose circumstances improve does not become non-violent in the same month.

A second, coarser form of learning is the escalation channel: having crossed the threshold makes crossing it again easier. It is scoped to the relationship, not to the man, and is undone when the relationship is released.

A.2.7 Prediction

None. No agent forms an expectation about the future, and no agent anticipates the consequences of acting. Deterrence is a raised threshold, the expected sanction represented as a parameter.

A.2.8 Sensing

Agents sense the following without error, cost, or delay:

- their own state, in full
- their partner's disposition, for men only and one-directionally (the dyadic channel runs from her state to his and not back)
- the disposition of each of the $|N|$ peers in their network, with the weights of that network
- for women, the annual action counters of their peers (who reported, who left, who stayed)
- the homicide rate of their own municipality in the current year
- their own household attributes, and the size of the relationship they belong to.

Agents do not sense the aggregate rates of anything, do not sense the state of the protective-order system, and do not sense agents outside their own network and relationship. A woman does not sense whether an order has been granted against her partner or whether a patrol has visited.

A.2.9 Interaction

Interaction is of three kinds and is otherwise absent.

Dyadic and direct. An assault is the only interaction with a physical consequence: it increments the victim's counters, sets her decision flag, may kill her, and erodes the aggressor's inhibition. Two interactions run in the other direction, from her to him. Through the dyadic channel her affective state enters his disposition target, and when she draws Remain his accumulated stress is reduced by a factor κ.

Networked and mediated. Peers enter one another's disposition through a weighted mean that is small against the agent's own disposition, so the network term damps disposition (§A.3.3, *Peer influence*). Among women only, they also influence one another through the annual action norm. Networks are same-gender by construction, so male and female norms propagate through separate structures. That choice follows the practice catalogued by Seff (2021).

Mediated by the model. Partner formation matches single agents of opposite gender in a randomised order. Custody removes a man from the schedule entirely.

There is no interaction through space, through markets, or through any collective actor.

A.2.10 Stochasticity

The model is stochastic throughout, and the sources are the following. The census sampling builds the population and draws its attributes jointly. Each agent draws its thresholds τ and τ^A, its network membership and the network's weights. Each month a Poisson call selects the stress branch, and there are draws for natural death, for employment and wage changes, for the violence trigger, for femicide and for the composition between Report and Separate. The protective chain draws the request, the grant, patrol coverage at the moment of grant, and arrest and removal on a breach. The remaining draws are post-separation contact, partner re-formation and the pairing penalty applied to black women.

Two properties of the stochastic design matter for reading the results. The violence trigger is a *logistic* hazard rather than a deterministic comparison, and the limit $s \to 0$ recovers the deterministic rule exactly. Under a deterministic comparison evaluated monthly for twelve years, whether a man is *ever* violent depends only on the lifetime maximum of his disposition, and maxima of long noisy series are nearly homogeneous across agents, so the socioeconomic gradient is discarded at the last step. The smooth hazard makes the whole trajectory count.

Second, the model is reproducible from a seed. Every collection iterated inside a draw is sorted by agent identifier and then shuffled with the model's own seeded generator, because iterating a Python set orders by memory address, which differs between processes and between runs. Without that discipline, common random numbers across scenarios are impossible. Both the NumPy global generator and the standard-library one are reseeded per replicate, since the model draws from each.

A.2.11 Collectives

Two collectives exist, and both are represented explicitly.

The *relationship* is an entity in its own right, with its own identifier and its own lifetime, and it is the scope of accumulation: assaults received, eroded inhibition, and the record of an order held all belong to it and are cleared when it is released. The lethality of an assault therefore depends on the history of the particular pairing and not on a disposition the aggressor carries between partners.

The *peer network* is a fixed, weighted, same-gender set of $|N|$ agents assigned at initialisation and not rewired thereafter. It has no identity of its own and no behaviour: it is the set over which two averages are taken.

A.2.12 Observation

Data are collected at three levels.

At the *step* level, the model's own collector records the counts of women who reported, women assaulted, adult women, and the summed stress.

At the *year* level, the harness reads the victim-side annual moments on the denominator the survey uses, women assaulted during the year, dead women included, since a woman killed during the year was assaulted during it. It registers a panel by AREAP and by municipality. It then resets the annual counters.

At the *run* level, the harness reads lifetime prevalence, order-level counts and breach composition, femicide composition by prior order, the post-separation counters, and the arrest count.

The three denominators differ. The annual moments are rates *among assaulted women*, the AREAP panel carries rates over *all adult women*, and the funnel composes rates whose denominators come from different institutions.

Nothing in the observation apparatus feeds back into behaviour, with one exception already named: the annual reset of the peer counters is behavioural, because the peer action norm reads them.

A.3 Details

A.3.1 Initialisation

Initialisation is data-driven and stochastic, and every run begins from a different draw of the same empirical distribution.

1. **Sampling.** The metropolitan region fixes a set of municipal codes and through them a set of AREAPs. A number of individuals equal to the requested number of relationships times the average household size of those AREAPs is drawn from the census age×gender×AREAP table, with probability proportional to the cell population.

2. **Attributes.** Each sampled individual receives years of schooling from the AREAP-level qualification distribution, constrained to be attainable at its age, a race category from the AREAP-level distribution and a wage drawn from a normal with the AREAP's own mean

and variance, then min–max normalised to $[0, 1]$ across the population. Employment status is drawn from a single population share, and work modality (formal or informal) uniformly among the employed.

3. **Pairing.** Within each AREAP, adult men and adult women are matched at random into relationships, one man per relationship. Individuals aged 18 or under are generated but not placed into relationships and do not enter the model. Where an AREAP has more women than men, the surplus women are not placed. Where it has more men, each surplus man forms a relationship of one.

4. **Agents.** Each paired individual becomes a *Men* or a *Woman* agent, receives an inhibition threshold $\tau_i \sim \mathcal{U}[\tau_{\min}, \tau_{\max}]$ and a birth month. A woman also receives an action threshold $\tau_i^A \sim \mathcal{U}[\tau_{\min}^A, \tau_{\max}^A]$. Both stocks, σ and δ (and δ^A), start at zero.

5. **Networks.** Once the whole population exists, each agent with at least $|N|$ same-gender others available is assigned $|N|$ of them without replacement, with weights drawn uniformly from $[w_{\min}, w_{\max}]$. Networks are assigned once and never rewired.

Implementation note (burn-in). There is none. Both stocks start at zero and climb towards their agent-specific equilibria under Rescorla–Wagner updating, so the first year or so of any run is a transient in which violence is below its stationary rate. The calibration moments are read over the whole 144-step horizon, so the transient is inside every calibrated quantity rather than excluded from it. Because every scenario shares the same transient and every contrast is a within-replicate paired difference under common random numbers, the transient cancels out of the differences the paper reports. It does not cancel out of the levels.

Implementation note (pairing and race). A black woman is matched to a proposed partner with probability $1 - 0.092$ rather than with certainty, which renders a documented differential in partnership rates. The consequence within the model is that black women are slightly under-represented among the partnered, and therefore among those exposed to the violence the model generates.

A.3.2 Input data

The model reads external data at two points: at initialisation, and once per simulated year.

Census microdata (IBGE, 2010). Population counts by age, gender and AREAP, the qualification distribution by AREAP, the race distribution by AREAP and the mean and variance of household wage by AREAP. These fix the joint distribution of agent attributes, so heterogeneity in the population is empirical rather than parametric, and every aggregate can be disaggregated back to the geography it was drawn from.

Municipal homicide series (DATASUS/SIM). Absolute homicide counts by municipality and year, divided by the census population of the municipality to give a rate per 100,000. This is read once per simulated year and is the only time-varying exogenous driver in the model. It ends in 2021, which fixes the horizon.

Attitudinal brackets (Fundação Perseu Abramo). The bracket scores by age band and schooling band from which the machismo index is assembled (Venturi and Godinho, 2013).

Institutional rates entered as constants. The share of reports carrying a request for an order (Civil Police of Rio Grande do Sul, Porto Alegre occurrence microdata 2012–2023) and the share of requests granted (Oliveira and Aguiar, 2025). Neither is adjusted to improve fit.

The model has no other exogenous drivers: there are no scenarios over the economy, no demographic projection, and no policy trajectory. Everything else is either a parameter or generated.

A.3.3 Submodels

Throughout, i indexes agents and t months. Parameter values are collected in Table A.4.

A.3.3.1 Natural death

At the start of each step an agent dies with probability

$$\Pr(\text{death}_i) = \frac{1}{M}\left[1 - \frac{\ln\big(|L_{g(i)} - \text{age}_i| + 1\big)}{\ln 101}\right], \tag{A.1}$$

clipped to $[0, 1]$, where L_g is a gender-specific life expectancy and M is a scaling constant. The hazard is a distance-to-life-expectancy rule rather than a life table: it peaks at the expectancy and falls away on both sides. The constant M is large, so mortality over the twelve-year horizon is low and the population is close to closed.

A.3.3.2 Stress accumulation

Stress is a stock. Each month a Poisson call at rate λ puts the agent on the accumulating branch with probability p and on the relief branch otherwise:

$$\sigma_i(t+1) = \begin{cases} \sigma_i(t) + s_i(t), & \text{with prob. } p, \\ \sigma_i(t)\,(1-\rho), & \text{with prob. } 1-p. \end{cases} \tag{A.2}$$

Forgetting is proportional to the stock rather than subtractive, so the stock cannot go negative and each agent settles at an equilibrium proportional to its own inflow, $\sigma_i^* = p\,s_i/[(1-p)\rho]$. Covariates therefore fix both the speed of accumulation and the *level* at which each agent comes to rest, so agents differ in disposition and not only in timing.

The inflow is the mean of ten additive components. Writing $d_i = \max\big(0, (Y_{\max} - y_i)/Y_{\max}\big)$ for the schooling deficit and using three qualitative severities HIGH > MEDIUM > LOW:

$$s_i(t) = \frac{1}{S}\Big[d_i\,(1+\omega_i) + g_i + (1-w_i)\,\text{HIGH} + h_i + n_i\,\text{HIGH} + a_i + \eta_i\,\text{LOW}\Big], \tag{A.3}$$

where ω_i weights the schooling deficit HIGH below six years of study and LOW at or above it. g_i is a gender constant, larger for women than for men. w_i is the individual wage and n_i the size of the relationship. a_i weights age HIGH between 19 and 28, MEDIUM to 59 and LOW from 60. η_i

is the count of assaults the agent has committed, which is a lifetime counter and is never reset. The household block is

$$h_i = 1 + (1 - \mu_i)\,\textsc{medium}, \tag{A.4}$$

with μ_i the work modality, the term entering only for agents who have one, so that informal work contributes more pressure than formal. Two further adjustments are applied to the partial sum inside the bracket: it is multiplied by $(1 + \textsc{low})$ when the agent's partner is black, and LOW is subtracted when the agent is employed.

Income enters the inflow once, through the individual wage term $(1 - w_i)\,\textsc{high}$. The constant in Equation (A.4) comes from a vestigial field (§A.4.2).

A.3.3.3 The societal channel

Two components, each normalised on its own scale before being mixed, so that the weight and not the units of the two sets their balance:

$$c_i(t) = w\,\tilde{m}_i + (1 - w)\,\tilde{v}_i(t). \tag{A.5}$$

$\tilde{m}_i$ is a machismo index built for male agents by scoring their age band and schooling band against the Perseu Abramo brackets, on two sub-scales ("common" and "critical" attitudes) that are each normalised and summed. The result runs on $[0.785, 1.488]$ for an adult and is min–max normalised against those bounds. It describes the aggressor's norms, so women carry zero on this component. $\tilde{v}_i(t)$ is the homicide rate of the agent's municipality in the current simulated year, divided by a reference rate and clipped at one. It is the only source of exogenous territorial variation: two otherwise identical men in different municipalities face different societal pressure.

A.3.3.4 Peer influence

The peer term on disposition is a weighted mean of the peers' dispositions minus the agent's own:

$$\nu_i(t) = \frac{1}{|N_i|} \sum_{j \in N_i} w_{ij}\,\delta_j(t) \;-\; \delta_i(t). \tag{A.6}$$

The weights w_{ij} are drawn from $[w_{\min}, w_{\max}]$ (Table A.4), at most 0.032, so the peers' part of the term averages about 0.011 against an own disposition of about 0.66. In a Porto Alegre run no agent had $\nu_i > 0$ at months 12, 72 or 144. The network channel therefore lowers the disposition target, by $\pi_N \tanh(\nu_i) \approx -0.12$ on average (standard deviation 0.03), and by more the higher the agent's own disposition. The mean is used rather than the sum, so the term does not grow with network size.

Prosocial peer norms do not enter here. Seeing peers report or leave does not make a woman less agitated. It makes her more willing to act on the agitation she already has, which is a different stock and is modelled as one (Equation (A.14)).

A.3.3.5 Disposition

A target is assembled from four channels, each min–max normalised into a bounded range and clipped:

$$\delta_i^*(t) = \underbrace{\pi_I\,\widetilde{\sigma}_i(t)}_{\text{individual}} + \underbrace{\pi_M\,\widetilde{c}_i(t)}_{\text{societal}} + \underbrace{\pi_D\,\widetilde{\delta}_{p(i)}(t)}_{\text{dyadic}} + \underbrace{\pi_N\tanh\big(\nu_i(t)\big)}_{\text{network}}. \tag{A.7}$$

The individual channel has the largest ceiling, more than four times any other, and the network, societal and dyadic channels follow in that order. The three non-individual ceilings sum to 0.40, below $\tau_{\min}$, so the environment cannot by itself carry a man across his threshold. The tanh squashing bounds the network contribution to $[-\pi_N, +\pi_N]$, which together with the clipping of the other three keeps disposition from feeding back into itself without bound.

The dyadic term applies to men only and runs one way, from her state to his. Her disposition enters normalised by the sum of the other three ceilings and clipped to $[0, 1]$. The term does not switch off at separation. While a post-separation tie stands it takes its ceiling value π_D, decaying linearly to zero across the window, because the motive the Brazilian case literature reads in post-separation violence is non-acceptance of the ending.

The agent then moves towards the target by Rescorla–Wagner updating:

$$\delta_i(t+1) = \delta_i(t) + \alpha\big(\delta_i^*(t) - \delta_i(t)\big). \tag{A.8}$$

A.3.3.6 The violence trigger

A partnered adult man assaults his partner in month t with probability

$$\Pr\big(\text{assault}_i(t)\big) = \Lambda\left(\frac{\delta_i(t) - \tau_i}{s}\right), \qquad \Lambda(z) = \frac{1}{1+e^{-z}}, \tag{A.9}$$

with the argument clipped to $[-60, 60]$ so the exponential cannot overflow. Setting $s \to 0$ recovers the deterministic Agent Zero rule $\delta_i > \tau_i$ exactly.

An assault increments the aggressor's and the victim's counters, sets the victim's decision flag for this step, draws the femicide of Equation (A.12) and then erodes the aggressor's inhibition. The erosion comes after the draw, so that the first assault is not also the escalated one.

A.3.3.7 Escalation

Having crossed the threshold makes crossing it again easier:

$$\tau_i \leftarrow \tau_i - \varepsilon, \qquad \text{cumulatively at most } \varepsilon_{\max} \text{ within one relationship.} \tag{A.10}$$

The cap is fixed by argument at the width of the τ draw, so a violent relationship can carry a man across the whole spread of inhibition present in the population, and no further. Without a cap he offends every month for as long as the relationship lasts.

The erosion is a property of the relationship. It is restored in full when the relationship is released (at the close of the post-separation window, at a breakup that never held violence, or when the man enters a new relationship), so a man who re-partners starts from the inhibition

he was drawn with. It is held in a separate offset from the protective order's boost, so the two compose and unwind independently.

A.3.3.8 Separation and the post-separation window

A relationship that has held violence persists after cohabitation ends, for a window of T^{sep} months. This is the sense of "relationship" used throughout: it ends when the window closes, not when the couple stops living together. One that has held no violence ends at the breakup, and everything it accumulated is cleared immediately.

While the tie stands, a man with no current partner may approach the woman he separated from. Contact is drawn first and violence second:

$$\Pr\big(\text{contact at } k \text{ months after separation}\big) = \phi\left(1 - \frac{k-1}{T^{\text{sep}}}\right), \qquad k \le T^{\text{sep}}, \tag{A.11}$$

decaying linearly to zero at the end of the window. If contact occurs, the ordinary trigger of Equation (A.9) decides whether it turns physical. The order of the two draws follows the law: under the Maria da Penha regime it is the *approach* that an order forbids, so the contact is already the breach whether or not it becomes violent, and drawing it first puts the order's deterrence on the pursuit rather than only on cohabiting violence.

When the window closes, the tie is released on both sides and with it goes everything it carried: the eroded inhibition, the count of assaults received, and the record of an order held in it. A man who re-partners stops pursuing.

A.3.3.9 Femicide

Conditional on an assault, the probability that it is lethal scales linearly with the number of assaults the woman has already received in the current relationship:

$$\Pr(\text{femicide}) = \min\big(m_i\,\beta,\ 1\big), \tag{A.12}$$

where m_i is incremented before the draw, so the first assault already carries probability β. The cap is formally present and never binds at the calibrated β. The count m_i is carried through the post-separation window intact and is cleared only when the relationship is released, so leaving does not reset the hazard, in line with the evidence that separation raises risk (Campos and Colman, 2024).

A femicide removes the woman from the population and puts the man in the incarceration register for a term counted in years.

A.3.3.10 The victim's decision

A woman carries a second stock, the propensity to act, whose target combines three components, the victim-side counterparts of the aggressor's channels, with weights summing to one:

$$\delta_i^{A*}(t) = \underbrace{\pi_E\, e_i(t)}_{\text{experience}} + \underbrace{\pi_P\, q_i(t)}_{\text{peer norm}} + \underbrace{\pi_A\, \mathbf{1}[\text{working}]}_{\text{autonomy}}, \tag{A.13}$$

updated by Equation (A.8) with the same learning rate. Experience is accumulated victimisation in the current relationship, saturating at a reference count: $e_i(t) = \min\big(n_i^{\mathrm{recv}}(t)/E_{\mathrm{ref}},\, 1\big)$, where the counter is the same m_i that drives lethality. The peer norm is the share of her female network who acted rather than stayed, from annual counters:

$$q_i(t) = \frac{1}{2}\,\frac{1}{|N_i|}\sum_{j\in N_i} d_j + \frac{1}{2}\,\frac{\sum_{j\in N_i} b_j}{\sum_{j\in N_i}(b_j + r_j)}, \tag{A.14}$$

where $d_j = 1$ if peer j has reported at least once in the current year and 0 otherwise, and b_j and r_j are the separations and stays of peer j in the current year. The second term is 0 when no peer has separated or stayed this year, and $q_i = 0$ for a woman with no network.

The decision stage is evaluated *only* in a month in which an assault occurred, and only against a current partner or the man she left. It is gated on the incident because the cycle-of-violence literature locates help-seeking in a window shortly after the assault, before the conciliatory phase. Conditional on having been assaulted:

$$\mathrm{action}_i(t) = \begin{cases} \text{Remain}, & \delta_i^A(t) \le \tau_i^A, \\ \text{Report}, & \delta_i^A(t) > \tau_i^A \text{ with prob. } \psi, \\ \text{Separate}, & \delta_i^A(t) > \tau_i^A \text{ with prob. } 1-\psi. \end{cases} \tag{A.15}$$

Separate is on the menu only while there is something to leave: once she has gone, the whole propensity routes to the formal channel. Remain applies the honeymoon feedback, $\sigma \leftarrow \sigma(1-\kappa)$ on the partner's stock, which gives staying a short-run pacifying effect.

Report sets the annual flag her peers read and opens the protective-order chain.

A.3.3.11 The protective-order chain

The chain is a sequence of conditional stages, each a separate institution with its own coverage, each switchable on its own.

1. **Request.** A report carries a request with probability p_{req}. Request is a stage of its own because the grant rate is published against requests, not against reports.

2. **Grant.** Conditional on a request, an order is granted with probability p_{MPU}. A man already under an active order does not accumulate a second one. Issuance is instantaneous, which the data support: 57.6% of orders are granted on the day of the request and 88.3% within two days, both below the model's monthly resolution. The two courts covering the modelled regions differ on this, TJRS granting 76% same-day against TJMG's 49%, and neither difference is resolvable at a monthly step.

3. **What the order does.** It runs for d_{MPU} months on its own clock and acts through two channels with different institutions behind them. *Deterrence*: the judicial act raises the aggressor's threshold by $\Delta\tau_{\mathrm{MPU}}$ for as long as the order stands, breach having been a crime in its own right since Lei 13.641/2018. *Monitoring*: at the moment of grant the order is drawn as visited with probability c_{patrol}. A visited order multiplies the aggressor's

disposition *and* accumulated stress by $(1 - \ell_{\text{MPU}})$ at every step of its term. An unvisited order carries the threshold boost alone and expires on the same clock.

The two offsets are commensurable by construction: the order raises the same threshold that escalation lowers. An order must be worth more than the erosion caused by the assault that prompted it, or it leaves the man more disinhibited than he was before.

4. **Breach.** An order in force is breached through either of two conducts. A man still living with his partner breaches by assaulting her, drawn from Equation (A.9) against the raised threshold. A man already separated breaches by approaching her, which art. 24-A defines independently of whether the approach turns physical. Each granted order is counted as breached at most once, so the model's non-compliance rate is a share of *orders*, which is how the statistic is constructed in the sources.

5. **Enforcement.** A breach draws two independent outcomes. With probability p_{arrest} it ends in custody: the man leaves the schedule for d_{arrest} months and nothing of his runs, which is the model's only channel that interrupts a violent spell mid-course. With probability p_{viol}, evaluated only on the in-relationship conduct, the relationship ends and the man detaches from the household, which is the removal from the home the order itself provides for.

6. **Expiry.** After d_{MPU} months the boost is withdrawn from his threshold. Any escalation offset stays until the relationship is released.

Compliance is indexed to no risk profile anywhere in the specification. The concentration of non-compliance in high-risk offenders that the model produces comes out of the disposition dynamics: high-risk men accumulate disposition faster, so the suppression a visit imposes competes against a stronger rebuilding tendency in exactly those men.

A.3.3.12 Relationship formation and dissolution

A relationship dissolves when the woman draws Separate, when the removal provision fires on a breached order, or when a member dies. Single agents are re-matched each month with a Poisson-derived probability, to the first eligible opposite-gender single in a reproducibly randomised order.

There is no exogenous dissolution hazard, so a couple that never sees violence parts only by death. The pool of single agents available for re-pairing holds the surplus men of initialisation, the widowed, and people who have left violent relationships or been left by them. Every relationship is nonetheless created with an expected duration that no process reads (§A.4.2).

A.3.3.13 Parameters

Table A.4: Parameters, with provenance. "Specialist" denotes elicitation carried over from the VIDA model, "set, not fitted" a constant fixed by design and not fitted to any moment, "data" a value taken from a source or cited study and "calibrated" one of the five parameters fitted to a moment.

Symbol	Meaning	Value	Provenance
Networks			
$\lvert N\rvert$	peer network size	18	set, not fitted
$[w_{\min}, w_{\max}]$	peer link weight range	$[5.86\times10^{-4}, 3.22\times10^{-2}]$	set, not fitted
Stress			
λ	Poisson call rate	1	set, not fitted
ρ	decay on a relief step	0.35	set, not fitted
S	number of inflow components	10	structural
$Y_{\max}$	schooling ceiling (years)	17	data
HIGH, MEDIUM, LOW	qualitative severities	0.4248, 0.2451, 0.0353	specialist
g	gender constant (men, women)	0.270, 0.730	specialist
Societal			
w	machismo vs. community weight	0.5	by argument (midpoint)
	machismo attainable range	[0.785, 1.488]	data
	community reference rate	100 per 100k	data
Disposition and trigger			
$\pi_I, \pi_M, \pi_N, \pi_D$	channel ceilings	0.985, 0.137, 0.217, 0.05	set, not fitted
α	learning rate	0.25	set, not fitted
s	trigger slope	0.02	set on the lifetime scale
$[\tau_{\min}, \tau_{\max}]$	inhibition threshold range	[0.933, 1.083]	calibrated (centre), width by argument
Escalation and post-separation			
ε	erosion per assault	0.013	calibrated (window [0.013, 0.015])
$\varepsilon_{\max}$	cap within a relationship	0.15	by argument (the width of τ)
T^{sep}	post-separation window (months)	12	data (Campos and Colman, 2024)
ϕ	contact hazard, first month	0.15	not identified by the moments
Femicide			
β	per-assault lethality increment	2.1×10^{-5}	calibrated (femicide rate, by proportion)
Victim action			
π_E, π_P, π_A	action channel weights	0.5, 0.3, 0.2	specialist
E_{ref}	experience saturation count	12	set, not fitted
ψ	reporting share of the act	0.571	by ratio argument
$[\tau^A_{\min}, \tau^A_{\max}]$	action threshold range	[0.13, 0.65]	calibrated
κ	honeymoon calming factor	0.05	set, not fitted
Protective-order chain			

continued on the next page

Table A.4, continued

Symbol	Meaning	Value	Provenance
p_{req}	request given report	0.653	data (Rio Grande do Sul. Polícia Civil. Gabinete da Chefia de Polícia, Divisão de Planejamento e Coordenação, 2024)
p_{MPU}	grant given request	0.655	data (Oliveira and Aguiar, 2025)
d_{MPU}	order term (months)	12	assumption
c_{patrol}	patrol coverage of granted orders	0.75	single-site estimate (Instituto de Pesquisa Econômica Aplicada, 2019)
ℓ_{MPU}	monitoring intensity	0.30	assumption, *not identified*
$\Delta\tau_{\text{MPU}}$	threshold boost	0.09	calibrated (non-compliance)
p_{arrest}	arrest given breach	0.17	data (floor)
d_{arrest}	months in custody	3	statutory floor, art. 24-A
p_{viol}	relationship ends given breach	0.40	assumption
Demography and employment			
L_g	life expectancy (men, women)	72, 79	set, not fitted
M	natural-death scaling constant	628.7	set, not fitted
	work modality μ (formal, informal)	1.158, 0.159	set, not fitted
	share employed at initialisation	0.548	set, not fitted
	monthly probability of a change in employment	0.022	set, not fitted
	maximum wage change	±3%	set, not fitted
	pairing probability withheld from black women	0.092	set, not fitted
	re-pairing Poisson rate	0.483	set, not fitted
	removal after a femicide (years)	1.04	set, not fitted
Run			
T	simulated months	144	design (data horizon)
	initial relationships	1000	design, the calibration holds only at this scale

Sources for the protective-order parameters. The provenance column of Table A.4 is terse for the chain, and five of its entries need their basis stated. $p_{\text{req}} = 0.653$ is the share of recorded threat and bodily-harm occurrences carrying a request in the Porto Alegre microdata (Rio Grande do Sul. Polícia Civil. Gabinete da Chefia de Polícia, Divisão de Planejamento e Coordenação, 2024). It is a period average and not a steady state, the share rising from 45.0% in 2012 to 68.8% in 2020, so a model run against a later window would want a higher value. $p_{\text{arrest}} = 0.17$ is derived from the Patrulha Maria da Penha of Piracicaba/SP over 2017–2019

(NEV/USP): 46 arrests in flagrante annualised over 91 recorded breaches. It counts municipal-guard arrests only and is therefore a floor and an order-of-magnitude anchor rather than a fine target. $d_{\mathrm{MPU}} = 12$ months is an assumption: the Lei Maria da Penha sets no fixed term, and the CNJ data indicate the critical window is the first six. $p_{\mathrm{viol}} = 0.40$ is an assumption representing the removal from the home the order itself provides for, and is distinct from custody, which is p_{arrest}. $c_{\mathrm{patrol}} = 0.75$ is a practitioner estimate at a single site (Instituto de Pesquisa Econômica Aplicada, 2019, pp. 45, 56) and an optimistic ceiling. The scenarios bracket it at 0 and 1 rather than treating it as a point estimate.

Three parameters carry a further qualification. The threshold boost $\Delta\tau_{\mathrm{MPU}} = 0.09$ is the one parameter of the chain fitted to a moment, the non-compliance rate 0.21 ± 0.04 per granted order, and it is larger than the escalation step $\varepsilon = 0.013$: an order must more than undo the inhibition that the assault motivating it had eroded, or it leaves the man more disinhibited than he was before. The monitoring intensity $\ell_{\mathrm{MPU}} = 0.30$ is an assumption reported as *not identified*, its moment sweeping flat because 91–94% of breached orders are breached in the first month, before the channel has run once. And $d_{\mathrm{arrest}} = 3$ months is the lower bound of the statutory term for art. 24-A (Lei 13.641/2018: detention of three months to two years).

A.4 Implementation, calibration and experiments

A.4.1 Implementation

The code is archived at CoMSES Net together with this document. The model is written in Python 3.12 on the Mesa agent-based modelling framework (Kazil et al., 2020), pinned to version 2.2.4 exactly rather than to a range: the frozen calibration was produced on that version, later 2.x releases change the warning behaviour of agent placement, and Mesa 3 removed the scheduler API the model is written against. NumPy is held below 2.0 for the same reason.

Replicates are independent and are run across processes. Each worker builds its own model, runs it, and reduces it to plain records before returning, because a running model holds agent references that do not serialise. Both random generators are reseeded inside each worker. Without that, a forked worker would inherit the parent's stream and a cell would report one run repeated N times with zero dispersion. The start method is `fork` where the platform has it and `spawn` where it does not, and the correctness of `spawn` is guaranteed by every harness setting its own parameter vector inside the worker entry point rather than trusting what it inherited.

One run of 144 steps at the Porto Alegre scale costs roughly 80 seconds on one core, of which about 7 seconds build the population.

A.4.2 Vestigial fields

Vestigial here means a field the code still writes and no process that produces a result reads. The archived code carries three. Each either consumes draws from the random stream or contributes a constant to a live equation, so deleting it would move the model's numbers. A reimplementer working from Sections A.1 to A.3 alone would not reproduce the archived runs number for number without them.

1. `Relationship.family_wage`, initialised to zero and never assigned. Two terms of the stress inflow read it, so both are constants. What survives of them is the additive 1 in Equation (A.4).

2. `Relationship.estimated_duration`, drawn at every relationship formation from an exponential whose rate comes from a Poisson draw repeated until it is positive, and never compared against the calendar.

3. The spatial grid. The model instantiates a 40×40 toroidal grid and places every agent and every relationship on it at a random cell, two draws each. No process reads that position, and the grid carries no mechanism: neighbourhood, exposure, and the community violence term are all resolved through the AREAP code rather than through grid adjacency (§A.1.2).

None of the three enters any mechanism. The draws behind `estimated_duration` and the two behind each grid placement consume the random stream, so removing them would shift every subsequent draw in the run. The constant that `family_wage` contributes sits inside the sum that S divides, so removing it would lower every agent's stress inflow by $1/S$.

The practical instruction for a reimplementer is: write Equation (A.4) as it stands, and reproduce the discarded draws in the order the code makes them if the object is to match the archived runs rather than to rebuild the model. All three are scheduled for removal in the next version, together with the recalibration that removing them requires.

A.4.3 Calibration

Five parameters are fitted to a moment. The centre of the aggressor threshold is fitted against lifetime prevalence, the victim's action threshold range against the annual reporting interval and the separation rate, the protective-order boost against the non-compliance rate, the escalation step ε against assaults per victim-year within the window the non-compliance rate leaves open, and the femicide base hazard β against the femicide rate. Expected femicide is linear in β, so β is set by proportion rather than by search. Two further quantities, the width of the threshold and the reporting share ψ, are fixed by argument rather than fitted, because the moments locate them only within a range.

The calibration is partially identified by construction. Lifetime prevalence identifies the *position* of τ relative to the disposition distribution and says nothing about its width. The width is chosen instead on the trade-off it governs. If it is narrow, who becomes violent is read almost entirely off the covariates. If it is wide, unexplained variation between men swamps them. The recorded width holds both, at the cost of passing under half of the available schooling gradient through to the outcome.

The victim channel is *over*-identified: one free parameter against two constraints, which makes the pair a specification test rather than a fit. It is also not separable from the aggressor side, because the decision is taken per assault while both targets are annual and per woman, so the per-assault probability that reproduces a given annual rate depends on how often women are assaulted, which is an output of the aggressor side. Any change upstream invalidates this channel and requires recalibrating all five together.

Tolerance bands are taken from the dispersion of each moment's own source rather than from an imported statistical criterion. The stopping rule is a band plus a re-check, with a ceiling of two passes.

A.4.4 Experimental design

The chain is evaluated by counterfactual scenarios. Each scenario overrides one or more parameters at run time and changes nothing else: the calibrated configuration is left untouched, the same population is instantiated from the same census substrate, and the scenarios run under common random numbers, so every contrast is a within-replicate paired difference and its standard error is the paired one. Each scenario is run for 150 replicates of 144 steps in each of the two metropolitan regions.

Table A.5: Policy scenarios and the parameters each overrides.

Scenario	Override	What it isolates
baseline	—	the calibrated configuration
no protective order	$p_{\text{req}} \to 0$	the whole chain: no report ever reaches an order
order granted but inert	$\Delta\tau_{\text{MPU}}, \ell_{\text{MPU}}, p_{\text{arrest}} \to 0$	the judicial *act* against its *execution*
no arrest on breach	$p_{\text{arrest}} \to 0$	the only channel that interrupts a spell mid-course
no patrol	$c_{\text{patrol}} \to 0$	removal of the patrol
full patrol	$c_{\text{patrol}} \to 1$	extension of the patrol to every granted order
reporting at the ceiling	$[\tau^A_{\min}, \tau^A_{\max}] \to [0.095, 0.475]$	the *upstream* intervention, encouraging assaulted women to act: reporting at the top of the survey band
no order and no patrol	both	consistency check, must reproduce "no order" digit for digit

The *inert* scenario leaves the removal provision standing, so a breach still ends the relationship with probability p_{viol}. Orders in that scenario are breached about three times as often as at baseline, so the residual channel fires far more often there than under the full order and not at all where there is no order to breach. "Inert" means inert as to deterrence, monitoring and arrest only. Neither p_{req} nor p_{MPU} is swept, because both are measured rates and a sweep over a measured rate traces a curve with no referent.

Sensitivity analysis sweeps the escalation step ε across five values at 40 replicates (SI B, *The escalation axis*).

A.5 Crosswalk to ODD+D

The ODD+D extension (Müller et al., 2013) adds elements specific to human decision making. Table A.6 maps each to where it is answered above.

Known simplifications

Collected here so that a reimplementer meets them in one place rather than discovering them one at a time.

Table A.6: ODD+D elements and where each is answered.

ODD+D element	Answer
Theoretical and empirical background	§A.2, "Basic principles": Heise's ecological framework, Epstein's Agent Zero, Rescorla–Wagner, the cycle-of-violence literature, and the enforcement literature, each at an identifiable place in the specification.
Individual decision making	§A.2 "Objectives" and §A.3.3.10. Two threshold crossings, no objective function, no comparison of alternatives. Remain is the default state.
Learning	§A.2 "Learning". Rescorla–Wagner on both stocks, plus escalation as within-relationship disinhibition. No learning about the institution.
Individual sensing	§A.2 "Sensing". Own state, partner's disposition (one way), peers' dispositions and annual acts, own municipality's homicide rate. Nothing aggregate, nothing about the order system.
Individual prediction	None. Deterrence is a raised threshold, not an anticipated sanction.
Interaction	§A.2 "Interaction". Dyadic (assault, dyadic channel, honeymoon), networked (the damping peer term on disposition, action norm), model-mediated (pairing, custody).
Collectives	§A.2 "Collectives". Relationship as an entity and the scope of accumulation. Peer network as a fixed neighbourhood. No institutional agents.
Heterogeneity	§A.3.1. The empirical joint distribution of the census, plus two independent uniform threshold draws. Heterogeneity is empirical rather than parametric.
Stochasticity	§A.2 "Stochasticity". Enumerated draws, the logistic trigger and why it is not deterministic, and the reproducibility discipline that makes common random numbers possible.
Observation	§A.2 "Observation". Three collection levels with three distinct denominators.
Implementation details	§A.4. Python 3.12, Mesa 2.2.4 pinned exactly, per-worker reseeding, paired scenarios.

- **Space is attributive, not relational.** Nothing is resolved through adjacency, and the grid the code instantiates carries no mechanism (§A.1.2, §A.4.2).
- **Income enters once, at the individual level.** Economic dependence reaches the model as a stress term for the aggressor and an autonomy indicator for the victim, but relative position within the couple not at all, which is the term Drigo et al. (2012) carries and where the bargaining accounts locate the mechanism (§A.3.3.2).
- **No exogenous relationship turnover.** Every separation is caused by violence, so the model cannot say what share of a city's separations are violence-related and it slightly overstates how concentrated a woman's exposure is in one partner. The calibrated moment is a rate among assaulted women and is unaffected (§A.3.3).
- **Three vestigial fields are retained in the code.** Grid coordinates, `family_wage` and `estimated_duration` affect no result but do affect the random stream (§A.4.2).
- **No burn-in.** Both stocks start at zero. The transient is inside the calibrated levels and cancels out of the paired contrasts (§A.3.1).

- **The peer term on disposition only damps.** At the drawn link weights the peers' weighted mean is about a hundredth of a disposition, so the term is negative for every agent and peers barely move one another's disposition (§A.3.3, *Peer influence*). Peers act on women's behaviour through the action norm.

- **The dyadic channel runs one way and is single-signed.** Katerndahl et al. (2022) find the partner's affective state signed by the specific emotion, "her anger maximized his violence, stalking, and alcohol use" while "her degree of upset hastened reduction of his violence" (p. NP20131). Here $\delta_{p(i)}$ is a single scalar with a positive weight and so carries only the escalating effect. Assaults received also do not feed the victim's own agitation, so his violence does not return to him through her disposition. Closing that loop would make its gain inseparable from the prevalence calibration.

- **Only formal action is counted.** The Report action is a police record, while among women who sought help at all only 1.9% went to formal sources against 31.3% who went to informal ones (Aboagye et al., 2023, p. 3), and which formal services women approach differs systematically across cultural groups (Satyen et al., 2019). The informal channel is folded into the peer norm rather than represented as an act, so a reporting rate calibrated on Brazilian data is not a structural constant of the model.

- **Social support is modelled in one direction only.** It raises the propensity to act. The same source is explicit that support "can also allow battered women to remain in violent relationships by reducing the negative impact" of the violence (Katerndahl et al., 2013, p. 1948), so the peer channel is biased towards action and the reporting rate it produces is an upper estimate.

- **No backlash spike.** Protection begins at the step the order is granted, with no short-term elevation of risk, although the CNJ timing data are consistent with one.

- **Children are generated and discarded.** They are sampled from the census and then excluded from relationships and do not enter the model.

- **One aggressor per relationship.** Violence is enacted by men against women in opposite-sex couples only.

SI B. Calibration targets, procedure and specification audit

SI B collects the material that supports the calibration of SobreVIDA without belonging to the argument of the paper. Section B.1 itemises what the model takes as given and what it produces, which fixes what may be read as a result. Section B.2 derives the six calibration moments from their sources and records the rule by which the calibration terminated. Section B.3 reports the diagnostics behind six specification choices, each of which was made against a measured alternative. Section B.4 gives the two held-out moments in full, and Section B.5 reads each scenario result in full. Section B.6 gives the calibration and the policy contrasts along the escalation axis. The model itself is described in SI A. Parameter values and their provenance are in its parameter table, Table A.4, which is the single authority for them and is not repeated here.

B.1 What is specified and what is generated

Table B.1 itemises what the model specifies and what it generates. Only the generated row is reported as a finding.

Table B.1: What the model takes as given and what it produces. Only the generated row contains results. The five free parameters of the calibrated row are reported with the calibration moments in the paper, *Calibration*. The fixed constants of the modeller-specified row are in the parameter table of SI A.

Status	What it covers	A result?
Specified by the modeller	Signs and functional forms of every stress and disposition term (SI A, *Submodels*). The empirical warrant for each is the warrant table, Table A.3. The Agent Zero trigger and its logistic hazard. The channel ceilings, scale factors, learning rate, calming factor κ, societal weight w and the reporting share ψ (SI A, *Parameters*).	No
Empirically given	The census joint distribution of agent attributes, the municipal homicide series, the machismo index (Venturi and Godinho, 2013), and the request and grant rates, entered as data and never adjusted to improve fit.	No
Calibrated to a moment	Five free parameters: the centre of the aggressor threshold (lifetime prevalence), the centre of the victim action threshold (the annual reporting interval and the separation rate), the protective-order boost (non-compliance rate), the escalation step (assaults per victim-year) and the femicide base hazard (femicide rate).	No
Generated	Level and dispersion of disposition, the share of men ever violent, the schooling gradient in violence, the spatial pattern of all of these and, once orders are active, the composition of femicide by prior MPU. The two held-out moments of the paper, *Out-of-sample tests* (repeat reporting within the year, against the Minas Gerais victim identifier, and the lifetime share no longer with the aggressor, against DataSenado) belong here, no parameter having been tuned against them. The *paired differences between scenarios.*	Yes

B.2 Calibration targets and procedure

The model is calibrated against six moments. The table of targets with their tolerance bands and the values obtained is in the paper, *Calibration*. The derivations that are not mechanical are as follows.

Lifetime prevalence is 0.267: the share of women reporting physical intimate-partner violence over their lifetime, unambiguously physical, partner, and lifetime (Fórum Brasileiro de Segurança Pública and Datafolha, 2025). *Reporting* is the interval $[0.142, 0.257]$ (Fórum Brasileiro de Segurança Pública and Datafolha, 2025): the floor is reporting to a women's police station, the ceiling is having sought any official body, and the partner-specific rate lies between because the report cross-tabulation is not published. *Assaults per victim-year* is the interval $[2.58, 3.55]$ (Fórum Brasileiro de Segurança Pública and Datafolha, 2025). *Non-compliance* is 0.21 ± 0.04, the year-on-year dispersion of the Porto Alegre occurrence file over 2019–2023. The two chain multipliers below reporting are exogenous measured constants (request rate 0.653 (Rio Grande do Sul. Polícia Civil. Gabinete da Chefia de Polícia, Divisão de Planejamento e Coordenação, 2024) and grant rate 0.655 (Oliveira and Aguiar, 2025)), not behaviours the model generates.

Separation enters twice, as a flow and as a stock, and only the flow is fitted. The calibration target is the annual rate, 0.15 ± 0.03. DataSenado corroborates it indirectly: 79% of women assaulted by a husband or partner are ex-partners at interview (Instituto de Pesquisa DataSenado, 2025, Tab. 1, p. 42), which is a stock rather than a flow, and a constant annual hazard h over T years accumulates to $1-(1-h)^T$, so $h = 0.15$ implies 0.80 at $T = 10$ against the 0.79 observed. That lifetime share is held out of calibration and tested in the paper, *Out-of-sample tests*. By the end of the twelve-year run the relationship with the aggressor has ended for 0.390 of ever-assaulted women in Porto Alegre and 0.373 in Belo Horizonte, and the women themselves left in 0.341 and 0.326. A victim is assaulted in 2.36 of the twelve years on average, so the flow accumulates over assaulted years and not over calendar years.

Table B.2: Separation by assaulted year. h_k is the share of women who leave in their k-th assaulted year among those who reached it without leaving. The cumulative share is $1-\prod_{j\le k}(1-h_j)$, and the constant column is $1-(1-0.161)^k$. Thirds of the action threshold τ_i^A are taken over all ever-assaulted women. Cells with fewer than 100 women at risk are blank. 150 replicates per city.

	Porto Alegre							Belo Horizonte	
			h_k by third of τ_i^A						
k	At risk	h_k	Low	Mid	High	Cumulative	Constant	h_k	Cumulative
1	32,329	0.053	0.146	0.014	0.001	0.053	0.161	0.053	0.053
2	18,870	0.162	0.311	0.150	0.054	0.207	0.296	0.156	0.201
3	11,238	0.278	0.439	0.324	0.153	0.427	0.409	0.267	0.414
4	5,632	0.347	0.516	0.459	0.238	0.626	0.504	0.318	0.600
5	2,379	0.331	0.586	0.485	0.252	0.750	0.584	0.338	0.735
6	937	0.251		0.465	0.208	0.813	0.651	0.278	0.809
7	404	0.235			0.214	0.857	0.707	0.201	0.847

Table B.2 is a life table over assaulted years. The hazard rises from 0.05 in the first assaulted year to 0.35 in the fourth, because accumulated victimisation is the largest component of the

propensity to act. In the first year it runs from 0.001 in the highest third of the action threshold to 0.146 in the lowest, and it rises within each third to the fifth year. The population hazard peaks in the fourth year in Porto Alegre and the fifth in Belo Horizonte and then falls, because by then the women with low thresholds have left. The cumulative share reaches 0.79 in the sixth assaulted year in both cities, against the ninth at a constant 0.161. At 2.36 assaulted years in twelve, six assaulted years take about thirty adult years, which extrapolates the hazard beyond the run.

Femicide is an anchor of scale, not a counted outcome. The Public Safety Yearbook gives 1.72 (2021) and 1.97 (2022) per 100,000 women in Rio Grande do Sul (Fórum Brasileiro de Segurança Pública, 2023, Table 27). Because the model's population is almost entirely exposed while the Yearbook's denominator is all women, four sourced factors, two on the numerator (partner or ex-partner authorship 0.798 and adult victims 0.955) and two on the denominator (adult female share 0.741 and ever-partnered share 0.807), convert this to a target of 2.19–2.51 per 100,000 in the model's denominator. The net factor is 1.27, because the numerator restrictions almost cancel the denominator restrictions.

Identification. Two of the calibrated thresholds convert a scale into a frequency. The aggressor's disposition and the victim's action disposition are constructed indices whose units belong to the model, and no survey measures "disposition to assault" in them. What the data identify is the *position* of a threshold within the realised distribution of its index, exactly as in a probit, where the error variance is fixed by convention because the latent utility carries no scale of its own and the observed frequencies pin down a position. Two scalars are asked to match three moments, so one is left over as a test, and a specification wrong in this part of the chain has no parameter left to absorb the mismatch. The test holds. The victim's single action threshold reaches the reporting rate and the separation rate at once.

Two further quantities are set by argument rather than fitted, because the moments locate them only within a range. The *width* of the aggressor's threshold is fixed on the trade-off it governs (Section B.3), and with the width fixed the centre is calibrated against lifetime prevalence, giving $\tau_i \sim \mathcal{U}[0.933, 1.083]$. And ψ, the composition of the victim's act, is computed from the ratio of the two annual targets rather than fitted to them, $\psi = 0.20/(0.20 + 0.15) = 0.571$, so it is not a free parameter. Revising either rate moves ψ. Composition and volume remain separate quantities under this rule, so the two annual rates are still free to disagree with the model, and at the calibrated action threshold $\mathcal{U}[0.13, 0.65]$ they do not. Two quantities are not identified by the available moments at all, the post-separation contact base and the monitoring intensity of a protective order, and they are carried into the results as assumptions.

The parameters of the protective-order chain. Nine parameters govern the chain and they divide three ways. Two are measured Brazilian rates entered as data: the share of reports carrying a request, p_{req}, from the Porto Alegre occurrence microdata, and the share of requests granted, p_{MPU}, from the CNJ. One is calibrated, the threshold boost $\Delta\tau_{\text{MPU}}$, fitted to the non-compliance moment and to nothing else. Its calibrated value exceeds the escalation step, so an order more than undoes the erosion of inhibition left by the assault that prompted it. The remaining six are assumptions with sources of varying strength: the order's term, patrol coverage,

monitoring intensity, the arrest probability and its term, and the removal probability. Of these six, the scenarios move c_{patrol}, ℓ_{MPU} and p_{arrest}. The other three hold their recorded values in every scenario, and the results are conditional on them. The monitoring intensity ℓ_{MPU} is *not identified*: non-compliance barely changes as ℓ_{MPU} is swept, because 91 to 94% of breached orders are breached in the first month, before the first visit. Values and sources are in Table A.4.

The shape behind the intensity mean. A level can be right while the distribution behind it is wrong. The intensity moment is a *mean*, assaults per victim-year, on target at 2.65, and the same source publishes the distribution that mean comes from. The escalation parameter was fitted against the mean and against none of the four shares, so the lower block of the table of targets in the paper, *Calibration* tests the shape with no parameter left to fit it.

The check fails, and in one direction. The model places 0.670 of victim-years in the lowest band against 0.519 observed, a gap of some seventy standard errors, and is short in all three bands above it. Violence in the model is spread too evenly across victim-years. The prevalence and intensity moments are unaffected, because they measure how much violence occurs and not how it is concentrated. No setting of the existing parameters closes the gap, because repetition within the year is governed by the escalation step alone and the same step governs breach, and raising it far enough to concentrate assaults carries non-compliance out of the administratively supported interval long before the shape is matched. Closing it requires a mechanism that produces repeated assault within a year *without* producing breach in the same proportion (the paper, *Limitations*).

How the calibration terminates. A chained calibration has no natural terminus: each block invalidates the one below, which on adjustment invalidates the one above. Four rules close the loop. *(i)* Each target carries a tolerance band, wide enough to cover the larger of the source's sampling error and the model's between-replicate noise. The first value inside the band is accepted and calibration stops there. *(ii)* Re-checking is a *test* and not a re-fit: after a block changes, one cell of the upstream block is re-run, and a value inside its band is recorded as re-checked with nothing moved. *(iii)* Passes are capped at two, exceeded once when the patrol mechanism was re-specified, which changes the specification rather than refining a fit. *(iv)* Calibration freezes by the calendar, and whatever is in the configuration at that point goes to the paper. Every target is therefore reported to the band the procedure resolved it to.

B.3 Specification audit

This section records the diagnostics behind six specification choices. Each reports why the specification takes the form it does, and is evidence about the design rather than a result about violence.

Why request is a stage of its own. The CNJ grant rate is conditional on a request having been filed: it counts orders conceded against orders requested. A chain running from report straight to a grant would apply that rate to the wrong denominator and issue roughly one and

a half times too many orders per report. The chain therefore separates request from grant and applies p_{MPU} downstream of p_{req} (SI A, *The protective-order chain*).

Why patrol coverage sits on the order. The Patrulha Maria da Penha visits only men under an order: every granted measure is forwarded to it and no visit occurs without one (Instituto de Pesquisa Econômica Aplicada, 2019, p. 56). A patrol channel firing for men *without* an order would reach the complement of that population, the 81% of violent men who never hold an order over the Porto Alegre baseline run, placing deterrence where the institution has none. Enforcement therefore sits on the order through c_{patrol}.

Why the trigger is a hazard and not a comparison. A deterministic rule $\delta_i(t) > \tau_i$, evaluated every month over the 144-step horizon, would make whether a man is *ever* violent depend only on the lifetime maximum of his disposition. Maxima of long noisy series are close to homogeneous: measured on this model, the standard deviation of the peak is 0.037 against 0.096 for the mean. Such a rule would discard at the last step the socioeconomic gradient the model exists to build: in a diagnostic run under it, schooling correlates -0.46 with mean disposition but only -0.07 with ever being violent, and the history-free rule $\mathbf{1}[\tau_i < \text{median peak}]$ reproduces 93.1 % of its violence classifications. The trigger instead uses a per-step logistic hazard of slope $s = 0.02$. At $T = 144$ and a constant gap, the lifetime probability of ever offending rises from 0.1 to 0.9 over a gap of about 0.06, wider than the standard deviation of the peak and comparable to that of τ (0.043).

The width of the inhibition threshold. Lifetime prevalence identifies the position of τ but not the width of its support, so the width was chosen on the trade-off it governs. Holding prevalence on target, widths of 0.10, 0.15 and 0.20 yield schooling gradients in violence of -0.20, -0.18 and -0.14 respectively, against a ceiling of approximately -0.40, which is the -0.52 gradient in mean disposition at the calibrated configuration attenuated by the dichotomisation a binary outcome implies at this base rate. We adopt 0.15, which retains unexplained heterogeneity while passing roughly 43 % of the achievable gradient through to the outcome. The centre is sensitive: at this width, displacing it by 0.05 moves prevalence from 0.315 to 0.089, so any change that shifts the disposition distribution requires recalibrating it, which is why this model calibrates in passes.

The resolution of the ψ grid. Two mechanisms should push the composition of the victim's act away from the ratio of the two targets: the share is drawn per *act* while the targets are per-woman flags, so the ≈ 1.25 acts per acting woman dilute both rates, and separation removes a woman from the pool that can act again. Both are real and neither is measurable at this sample size. Sweeping the composition across a 0.07-wide window moves the reporting rate by 0.018 and the separation rate by 0.025, while two independent replications of the same grid disagree with each other by 0.008 and 0.010 on average. The grid locates ψ to no better than about ± 0.02, an interval narrower than the range from 0.486 to 0.631 that the targets themselves imply, which is why ψ is fixed by argument rather than fitted.

Why the inert order moves separation in two directions. The separation rate under an inert protective order points in opposite directions according to what it is compared against. Against the baseline an inert order *raises* separations (+0.011, z^*=6.8 in Porto Alegre; +0.014, z^*=10.0 in Belo Horizonte), because an unenforced order leaves more violence for the woman to respond to. Against no order at all it *lowers* them (−0.033, z^*=−21.1; −0.027, z^*=−16.0), because exits she would have made herself are pre-empted by the provision that removes the aggressor from the home on breach, and the rate counts only the exits she takes.

B.4 The held-out moments

Too many reports within the year. The pseudonymous victim identifier in the Minas Gerais microdata makes repeat reports countable (Minas Gerais. Polícia Civil, 2024). Among women who report in a given year, the observed number of reports is 1.163. The model produces 1.376 in Belo Horizonte, an 18% excess (z=16), with roughly double the observed mass on "twice in the year". The reporting *rate* in the same 40 replicates stays inside its calibrated band (0.182, within $[0.142, 0.257]$), so the excess is in repeat reports and not in the level. Two features of the source push the comparison the same way: a single police occurrence may aggregate several episodes, and the target counts only three offence types, so the observed 1.163 is a floor.

The excess comes from one part of the specification. The victim's propensity to act moves each month towards a target that combines accumulated victimisation, peer norms and economic autonomy, and *her own act enters no term.* The peer component counts what *other* women did, never what she did. A woman keeps no memory of having reported, so once her propensity crosses her threshold she reports at every assault in the year. Raising her threshold would reduce the repeats and move the calibrated reporting and separation rates with them. The aggressor already keeps a memory of his own acts, through the erosion of his threshold (SI A, *Submodels*), and Katerndahl et al. (2021a,b) separate prior experience of acting from readiness and from the immediate need a given episode creates.

The lifetime separation stock. In the DataSenado survey, 79% of the women who report having been assaulted by a partner describe that partner as an ex-partner at the time of interview, a stock accumulated over however many years the respondent has been in relationships. In the model, the relationship with the aggressor has ended for 0.390 of ever-assaulted women by the end of the twelve-year run (standard error 0.003), by whatever route: she left, the removal provision of a breached order ended it, or he died. A share of 0.341 left through the action stage. The life table over assaulted years, Table B.2 in Section B.2, shows the cumulative share who left reaching 0.79 in the sixth assaulted year, which at 2.36 assaulted years in twelve takes about thirty adult years. The run is short because the municipal homicide series that drives the societal term ends in 2021, so the thirty years extrapolate a hazard measured within twelve.

Two further quantities held out, from the CNJ. A CNJ presentation reports the national stock of orders (Oliveira and Aguiar, 2025). Its grant rate $p_{\mathrm{MPU}} = 0.655$ enters the model as data. Two further quantities are held out. Of 5,144 femicide cases with identified victims in 2024–2025, 32.4% had an order previously granted against the same defendant. While 33% of

femicide cases were attempted rather than completed, among victims holding a prior order the share rises to 41%, a relative increase in survival of 24% conditional on escalation. The same source carries a pattern the model does not reproduce: 24.1% of those victims were killed in the same month as the grant and 49% within six months (the paper, *Limitations*).

B.5 The scenario results in detail

This section reads each result of the scenario table in the paper, *Counterfactual scenarios* in full. Entries are paired within-replicate differences against the baseline under common random numbers, 150 replicates, and only $|z^*| > 3$ counts as an effect.

R1. The protective chain does reduce violence, modestly. Dismantle it entirely, so that no order is ever issued, and women in the model are assaulted more often: assaults per woman-year rise 2.9% in Porto Alegre (z^*=3.3) and 5.9% in Belo Horizonte (z^*=6.6), and expected femicide rises by 4.3% and 4.8%, neither above the threshold. The chain reduces violence by a few per cent.

R2. Enforcement changes compliance far more than violence. The order can be taken apart. One scenario keeps the judicial act and strips it of everything that follows: the order is granted and recorded, but deterrence, monitoring and arrest on breach are all switched off. Aggressors under such an inert order breach it about three times as often as under a live one (non-compliance 0.244 → 0.699 in Porto Alegre and 0.231 → 0.697 in Belo Horizonte, both $z^* > 60$), and yet the violence experienced by women in the city hardly moves (+2.3% and +2.5%).

The less enforcement, the more non-compliance. Ranked by how much enforcement they leave standing, the four scenarios order non-compliance the same way in both cities: 0.178/0.170 under full patrol, 0.244/0.231 at baseline, 0.399/0.390 with patrol removed and 0.699/0.697 once the order is inert. Violence does not follow that order, so an agency can improve markedly on compliance while the violence it exists to reduce stays where it was.

The scenario table of the paper, *Counterfactual scenarios* separates the two halves of the order: the step from baseline to inert is what losing execution costs, and the step from inert to no order at all is what the judicial act is worth on its own. That second step is the one contrast the table does not carry, because it pairs two scenarios against each other rather than against the baseline. Run within replicate it is −0.0008 assaults per woman-year in Porto Alegre (z^*=−0.7), not distinguishable from simulation noise, and −0.0042 in Belo Horizonte (z^*=−4.2), which counts as an effect.

The Belo Horizonte step does not measure deterrence. The inert scenario strips deterrence, visiting and custody but leaves the removal provision standing (the paper, *Policy scenarios*). Because inert orders are breached three times as often, that provision fires far more often under the inert order than at baseline, and it cannot fire at all where no order exists to breach. Even an order that exists only on paper therefore ends some relationships when it is breached, and ending a relationship removes the woman from the setting in which escalation happens. The separation rate moves accordingly (Section B.3).

R3. The intervention with a large effect is upstream. The last intervention encourages an assaulted woman to act, so that she reports or leaves earlier, and the share reporting rises from the calibrated 0.185 to the top of the range the national survey observes. Assaults per woman-year fall by about a quarter and expected femicide by more than half, with $|z^*| > 31$ in both cities, against a few per cent for every scenario that works on enforcement. Like *full patrol*, which moves coverage from 0.75 to 1.0, it changes a single parameter, the woman's threshold to act.

The intervention works mainly through separation. It raises both acts: per 100 women assaulted in a year, reporting rises from 18.5 to 25.5 and separation from 16.1 to 21.6 in Porto Alegre, with the same increases, to the first decimal, in Belo Horizonte. Of the 12.6 additional women who act in Porto Alegre, 2.3 reach an enforced order. *Full patrol* enforces more orders than this intervention (8,309 against 7,422 in Porto Alegre, 7,693 against 7,204 in Belo Horizonte) and does not reduce violence, so the orders do not explain the effect. The number of assaults per victim falls, from 6.50 to 4.80 in Porto Alegre and from 6.17 to 4.58 in Belo Horizonte, while lifetime prevalence does not move. A woman who leaves earlier is spared later assaults, not the first one. Expected femicide falls more than assaults because the lethality hazard grows with the assaults accumulated in the relationship (the paper, *Escalation, separation and lethality*), and leaving earlier stops that count sooner.

R4. Withdrawing the patrol raises violence. Extending it does not reduce it. Taking patrol coverage from its current 0.75 to full does not reduce violence in either city (Porto Alegre +0.7%, z^*=1.0; Belo Horizonte +1.1%, z^*=1.7), even though it cuts non-compliance by about a quarter. Withdrawing the patrol that exists raises violence in Belo Horizonte (+2.8%, z^*=3.7). In Porto Alegre the difference is not distinguishable from noise (+0.7%, z^*=0.9).[1]

Full coverage does little because the patrol reaches few women, and reaches them late. It visits only men whose partner already holds a granted order, about 8 of every 100 assaulted women, and full coverage adds about 2 of those 100. And 91 to 94% of breaches happen in the first month of the order, before the first visit (Section B.2).

R5. No scenario changes lifetime prevalence. The share of women ever assaulted is nearly flat across every scenario (0.271 to 0.272 in Porto Alegre, 0.244 to 0.247 in Belo Horizonte), including the reporting intervention whose assaults fall by a quarter. All six scenarios act on women who have already been assaulted, so none can prevent a first assault. They reduce how often the violence recurs and how likely it is to end in death. Prevalence is therefore the wrong statistic against which to evaluate them. Primary prevention, acting before any violence, was not among the interventions tested.

R6. Reporting barely responds to enforcement. The three downstream rates can be held fixed while enforcement varies only if reporting does not respond to enforcement. Removing the order, making it inert, removing arrest on breach or taking patrol to full coverage all leave the reporting rate within 4% of where it started. The largest change comes with *no patrol* (+3.2%,

[1] Along the escalation axis (Section B.6), the Porto Alegre estimate for removing patrol is positive at four of the five values of the escalation step, though at 40 replicates none reaches $|z^*| = 3$.

z^*= + 3.1 in Porto Alegre; +4.1%, z^*= + 3.8 in Belo Horizonte): withdrawing enforcement produces more violence, and a few more women act as a result. *Full patrol* does not move reporting (+0.0%; −1.2%). The upstream intervention raises reporting by roughly 38% and 39% ($z^* > 31$). In the model an enforcement scenario reaches reporting only through the violence it changes (the paper, *Limitations*).

The femicide level. Expected femicide is 2.33 per 100,000 in Porto Alegre, inside the target band $[2.19, 2.51]$, and 2.03 in Belo Horizonte, about 7% below the lower bound. It is predicted rather than counted: at roughly 0.2 deaths per run counting would be noise, so we report FEMICIDE_BASE $\times \sum n_{\text{prior}}$ over the woman-year panel (Section B.2).

B.6 The escalation axis

The escalation step ε is swept across $[0.010, 0.030]$, roughly tripling the parameter and travelling far beyond the window the calibration resolves it to, and every calibrated moment is re-measured at each value. In Table B.3 the recorded value is the only one of the five to satisfy all six bands at once. The calibration fits one moment at a time, and the sweep re-measures all six on the frozen configuration. Prevalence is flat across the whole axis (0.271 to 0.276 over a parameter that triples), so escalation changes how violence is distributed over women already assaulted and not how many are assaulted.

Table B.3: Calibration moments along the escalation step. Each cell is a calibrated moment, and ✓ marks a value inside its tolerance band, ✗ outside. Only the recorded value (0.013) passes all six moments at once. Prevalence is flat across the whole axis. Porto Alegre, 40 replicates. Source: authors' elaboration on model output.

Step	Prevalence	Intensity	Reporting	Separation	Non-compliance	Femicide
0.010	0.271 ✓	2.392 ✗	0.167 ✓	0.140 ✓	0.173 ✓	1.90 ✗
0.013	**0.275** ✓	**2.627** ✓	**0.185** ✓	**0.164** ✓	**0.247** ✓	**2.27** ✓
0.020	0.276 ✓	3.039 ✓	0.221 ✓	0.190 ✗	0.297 ✗	3.05 ✗
0.025	0.274 ✓	3.247 ✓	0.245 ✓	0.210 ✗	0.304 ✗	3.35 ✗
0.030	0.274 ✓	3.416 ✓	0.251 ✓	0.224 ✗	0.325 ✗	3.71 ✗

The policy contrasts along the axis. Read as paired deltas on assaults per woman-year (Table B.4), extending patrol coverage from 0.75 to 1.0 moves violence at *no* value of the axis ($|z^*| \leq 1.3$ throughout, including where four of the six calibration moments fail). Removing the order chain (*no protective order*) gives a positive estimate at all five values, three of them above $|z^*| = 3$ (z^* from +1.5 to +5.4). Removing patrol gives a positive estimate at four of the five values, none of them above $|z^*| = 3$ (the largest is z^*= + 2.7).

The funnel along the axis. Because only the entry rate is a model output, the funnel follows the reporting rate along the axis, from 16.7 → 10.9 → 7.2 → 5.4 per 100 at the low end to 25.1 → 16.4 → 10.8 → 8.1 at the high end. Tripling the step moves the enforcement ceiling from

Table B.4: Policy contrasts along the escalation axis. Paired CRN deltas on assaults per woman-year (z^* in parentheses), Porto Alegre, 40 replicates. *Full patrol* is null throughout and *no order* is positive throughout. Source: authors' elaboration on model output.

Step	No patrol	Full patrol	No order
0.010	+3.0% (+2.0)	+0.3% (+0.2)	+8.2% (+5.4)
0.013	−0.6% (−0.4)	+0.1% (+0.1)	+5.3% (+3.2)
0.020	+2.9% (+2.5)	+0.1% (+0.1)	+2.4% (+1.5)
0.025	+0.4% (+0.3)	−1.8% (−1.3)	+5.4% (+3.8)
0.030	+3.4% (+2.7)	−0.6% (−0.4)	+3.1% (+2.1)

7.2% to 10.8%, against 7.9% at the calibrated value. At every value, most assaulted women are lost at the first stage, reporting.[2]

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[2]This axis is run at 40 replicates, against the 150 of the scenario table in the paper, *Counterfactual scenarios*. Two of its results hold at that count: the *full patrol* null ($|z^*| \leq 1.3$ at every value), and the ceiling interval, which is arithmetic on the reporting rate. The magnitude of *no protective order* is less precise. At the calibrated value this axis gives +5.3% assaults per woman-year with a standard error of 1.7 percentage points, against +2.9% in that table, the same sign and a gap of about 1.4 standard errors. We quote the 150-replicate figure wherever a magnitude is given.

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